

Siddharth Ravani

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EDUCATION

Sri Sai Ram Engineering College

Bachelor of Engineering in Mechanical Engineering (CGPA: 8.44)

Chennai, TN

2023 – 2027

Everwin Matriculation Higher Secondary School

Grade- 87.5%

Chennai, TN

2021 – 2023

PROFESSIONAL EXPERIENCE

Industrial Manufacturing Intern

Microtech Engineers

January 3 – January 17, 2025

Chennai, TN

- Worked closely with a team of experienced engineers to understand end-to-end manufacturing processes, from tool design to CNC machining and mould testing.
- Gained firsthand exposure to plastic injection molding operations, including material loading, die setup, and mold cycle optimization.
- Developed a working knowledge of industrial safety practices, shop floor protocols, and how CAD models are translated into physical products.

Engine Management and Fuel Systems Intern

Delphi-TVS — 15-Day Internship

December 2025 – January 2026

Chennai, TN

- Developed a clear understanding of ECU calibration processes used to meet BS6 emission standards.
- Gained knowledge of emission testing methods and validation procedures followed during ECU calibration.
- Acquired hands-on exposure to Common Rail (CR) fuel pump assembly and disassembly, understanding component functions.
- Learned about CR fuel pump validation tests to assess performance, durability, and reliability
- Studied CR pump classifications and pump family types used across different engine applications.

MECHANICAL DESIGN PROJECTS

7-Segment Mechanical Clock | *Cam-Actuated 7-Segment Mechanical Clock.*

- Designed a fully mechanical 7-segment display clock where each segment is physically actuated by custom-profiled cams, eliminating the need for electronics.
- Modeled all individual components — cams, followers, segment arms, and housing — using SolidWorks 3D part modeling with precise geometry and tolerancing.
- Assembled the mechanism using SolidWorks mates and constraints to ensure accurate cam-follower contact and synchronized segment movement.

Theo Jansen–Based Horse Walking Mechanism | *A SolidWorks project exploring the Theo Jansen mechanism.*

- Designed a multi-link legged locomotion mechanism based on the Theo Jansen linkage to convert rotary input into stable walking motion.
- Tools: Adjustable Wrench, Allen Key, ER Collet
- Modeled and assembled the complete mechanism in SolidWorks, refining link dimensions through iterative motion studies.

WEB DEVELOPMENT PROJECTS

Razorpay Clone | *HTML/Tailwind CSS/JavaScript*

- Responsive landing page UI developed using Tailwind CSS, mimicking the real Razorpay homepage.

Mo.co Website Clone | *HTML/CSS*

- Website replica focusing on layout precision and CSS styling.

TECHNICAL SKILLS

CAD/Design Tools: SolidWorks(CSWA), Fusion 360, AutoCAD

Programming Languages: Embedded C/C++, HTML/CSS, JavaScript

Frameworks: React, Node.js